
WELCOME

OBJECTIVES

- Introducing the students to the basic semiconductor devices, their characteristics and major applications.
- Getting familiar with datasheets of some devices used in the lab and looking a practical aspects and limitations related to them.
- Exposing the students to basic laboratory instruments, circuit boards, and modern engineering tools
- Learning some basic circuit-design skills related to semiconductor devices applications and their biasing
- Practicing technical writing skills

EVALUATION AND GRADING POLICY

Attendance	5%
Reports/Mini Reports	20%
Assignments	5%
Midterm Exam	20%
Design and Simulation Quiz	10%
Final Exam	40%

General Rules To Be Observed in The Lab

- Foods and drinks Are not allowed in the lab.
- Do not use mobile phone except for Emergency.
- Do not relocate any instrument or equipment. Ask the instructor if you need help.
- Before you leave the lab, switch power sources off, put your components back, and keep your bench tidy and clean.

Reports Guidelines

- Guidelines given on Page 5 and 6 of the lab manual
- Pay attention to the **observation and discussion** parts
- Must be submitted before deadlines via LMS
- Usual due time is 11:30 pm
- Late reports/assignments are not accepted
- You are not allowed to submit a report for an experiment that you did not attend

EE 312
Exp 1

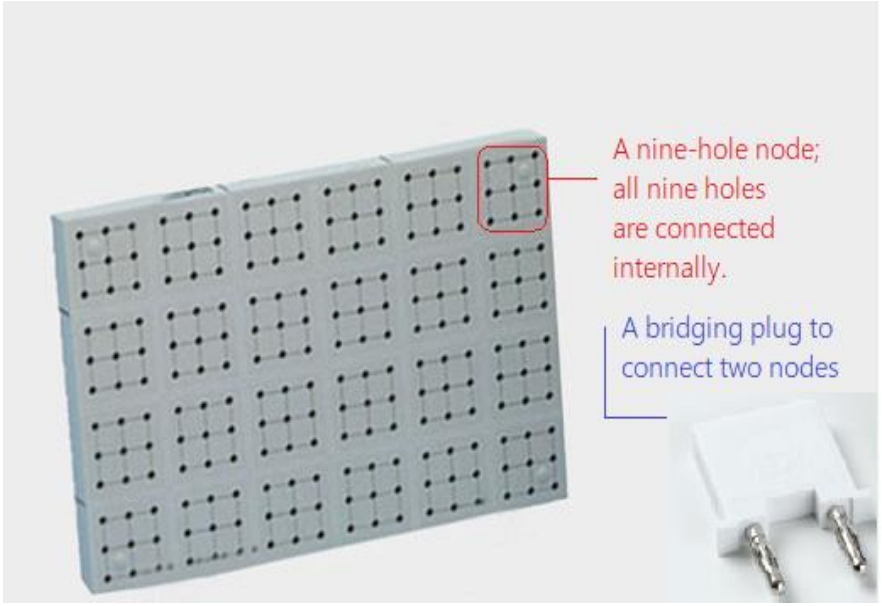
INTRODUCTION TO PN JUNCTION DIODES

IN THIS EXPERIMENT

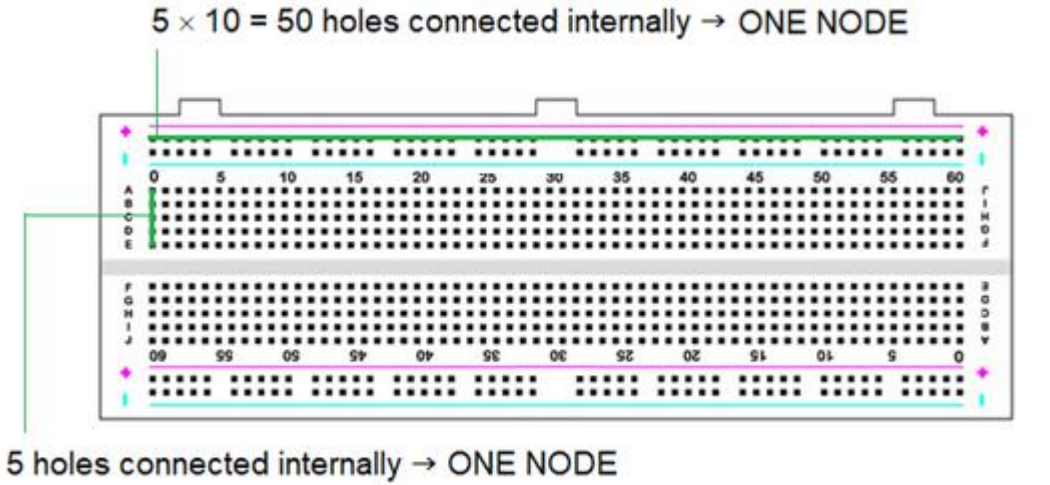
- Four types of diodes are introduced.
- Testing of diodes by digital multimeter is illustrated.
- The behavior of the diode will be explored for the:
  - forward bias case.
  - reverse bias case.
- The voltage-reference application will be examined.

BORADS USED IN THIS LAB

Socket Board



Breadborad



There are 61+61 = 122 five-hole vertical nodes *and* 4 fifty-hole nodes

TEST LEADS TYPES



BANANA PLUG to BANANA TEST LEADS



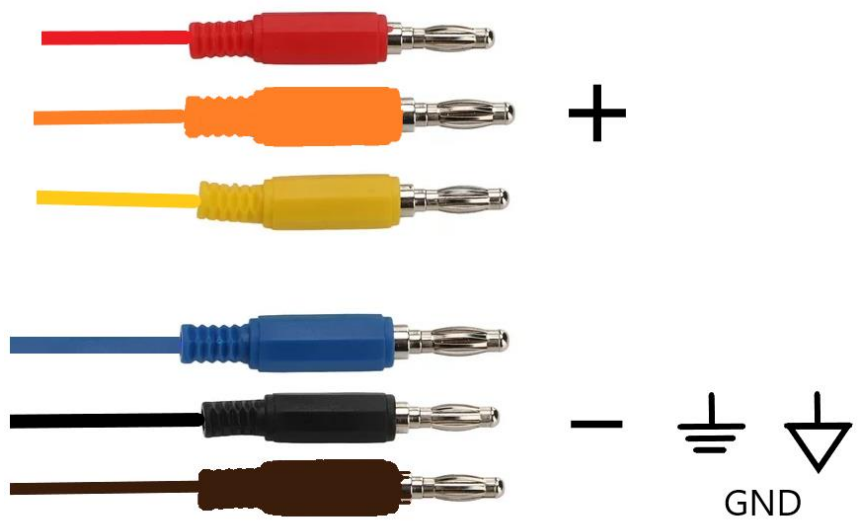
MINI GRABBER HOOK to BANANA TEST LEADS



ALLIGATOR CLIP to BANANA TEST LEADS

Using Right Colors

- Makes your Circuit Look Professional
- Reduces Probability of Wrongful Connections
- Makes Maintenance Easy



THE DIGITAL MULTIMETER (DMM)

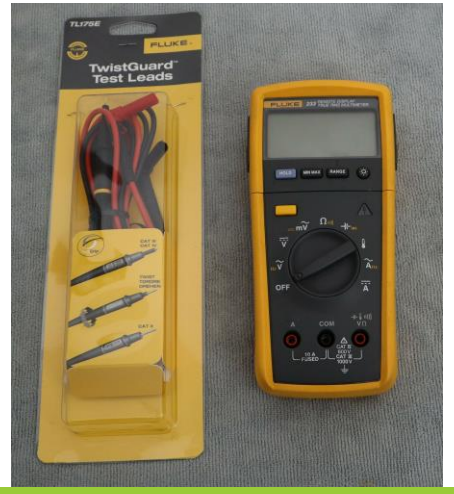
Very important instrument which engineers use frequently for measurement and testing.

It can measure: resistance, ac and dc current, ac and dc voltage, and in some types it can also measure capacitance and frequency.

It can test wires, diodes, and transistors.

It can be hand-held or bench-type.





MAIN FUNCTIONS OF THE DMM



use these two jacks to measure voltage or resistance, and to test a wire or a diode

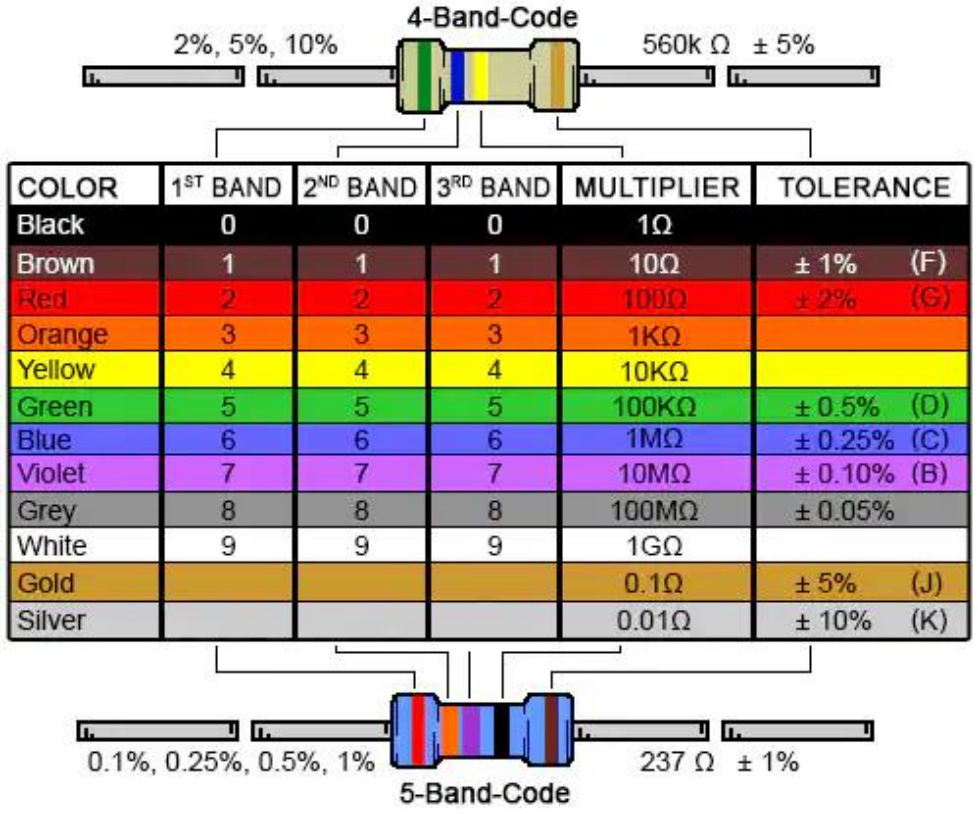
To test a wire, press this button

To test a diode, press this button

To measure resistance, press this button

this jack is used for current measurement, which we will not measure directly in this course; we will calculate current by dividing the measured voltge by the measured resistance

RESISTOR COLOR CODE



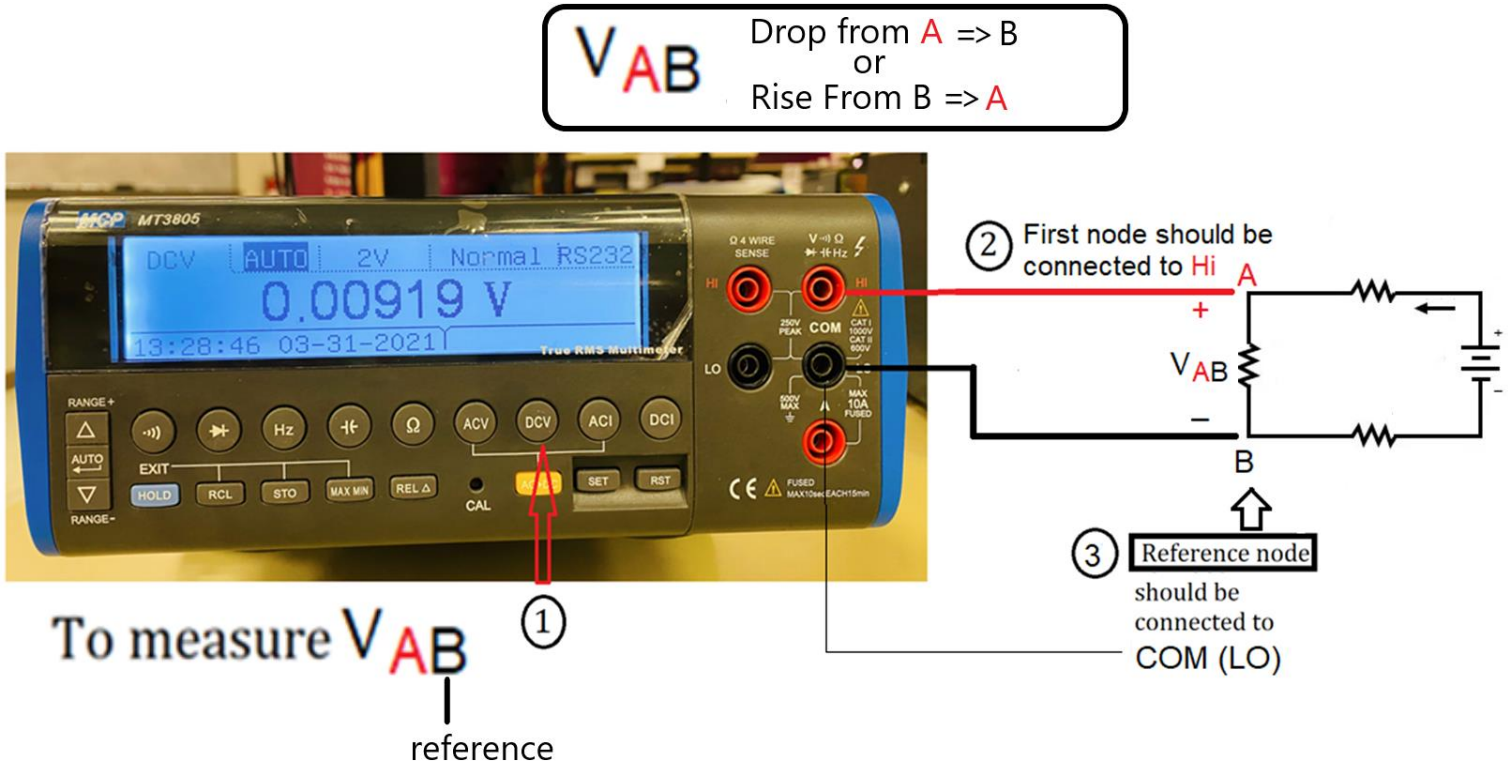
COLOR	1ST BAND	2ND BAND	3RD BAND	MULTIPLIER	TOLERANCE
Black	0	0	0	1Ω	
Brown	1	1	1	10Ω	± 1% (F)
Red	2	2	2	100Ω	± 2% (G)
Orange	3	3	3	1KΩ	
Yellow	4	4	4	10KΩ	
Green	5	5	5	100KΩ	± 0.5% (D)
Blue	6	6	6	1MΩ	± 0.25% (C)
Violet	7	7	7	10MΩ	± 0.10% (B)
Grey	8	8	8	100MΩ	± 0.05%
White	9	9	9	1GΩ	
Gold				0.1Ω	± 5% (J)
Silver				0.01Ω	± 10% (K)

HOW TO USE DMM TO MEASURE RESISTANCE

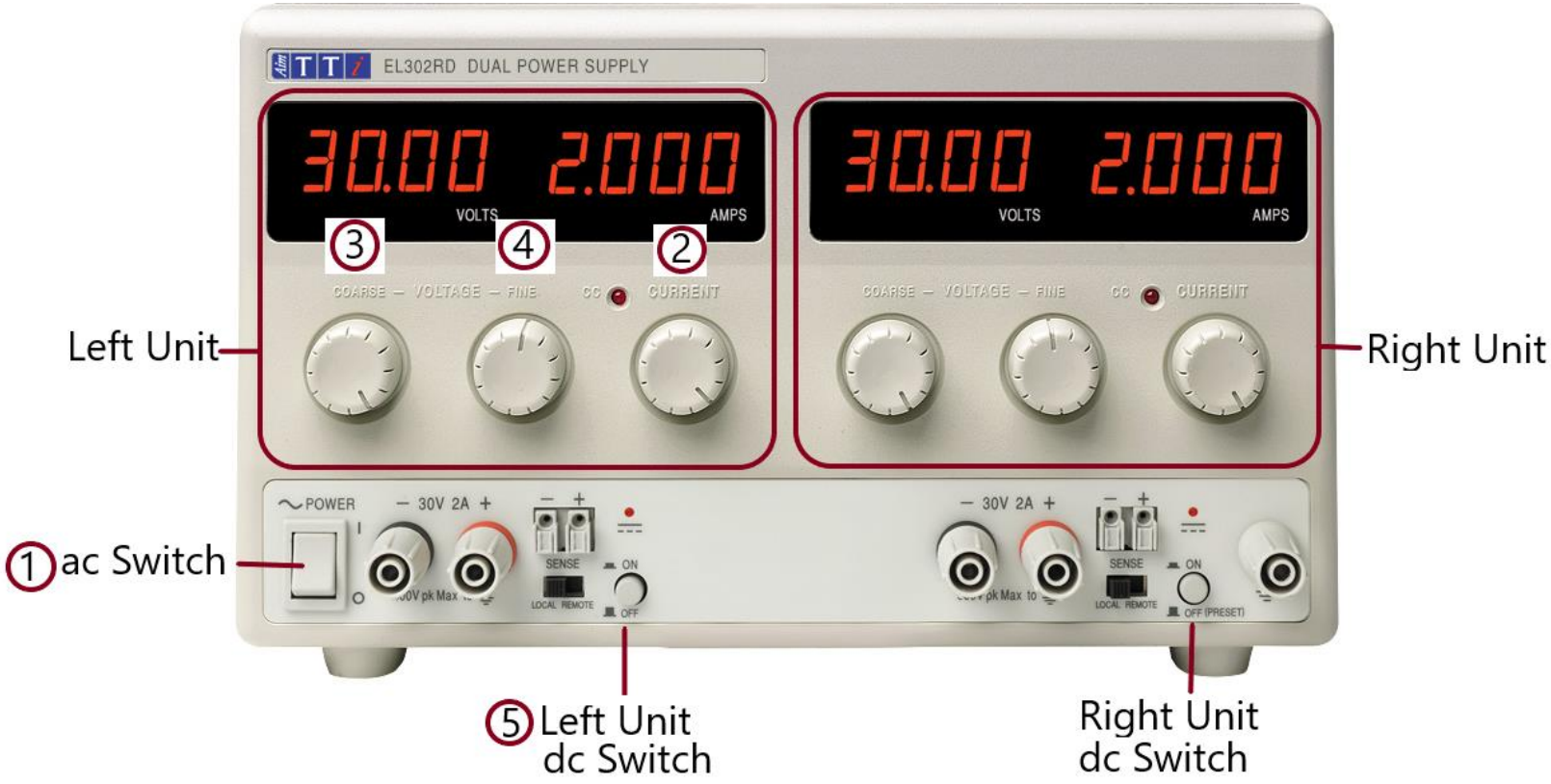




HOW TO USE DMM TO MEASURE DC VOLTAGE



THE DC POWER SUPPLY

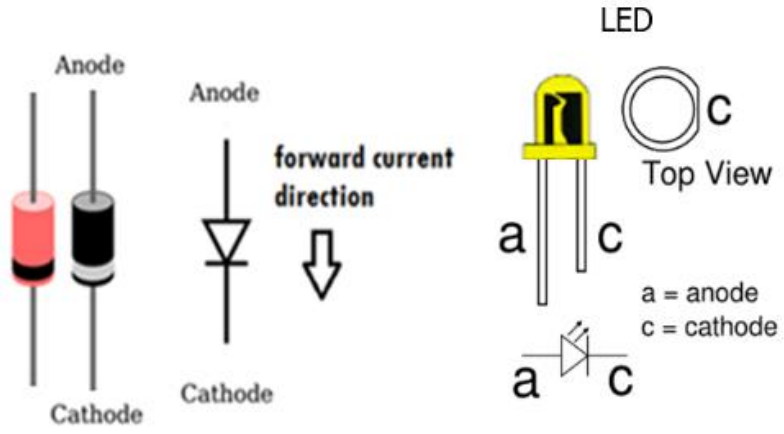


WHAT IS A P-N JUNCTION DIODE?

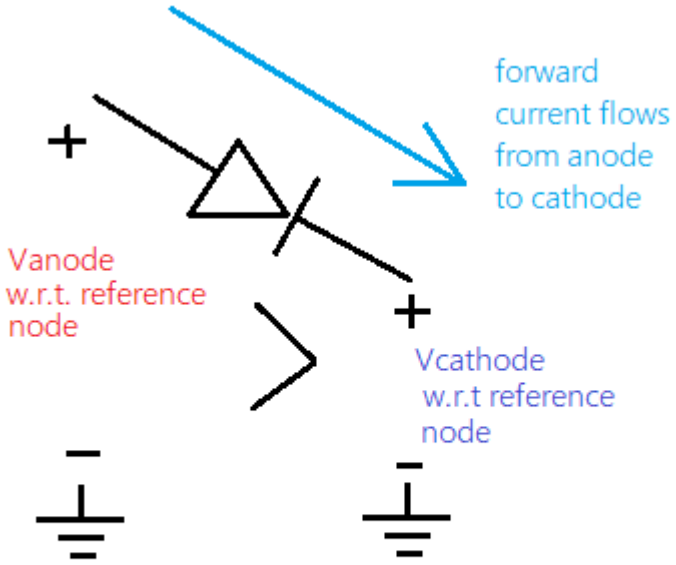
A p-n junction diode is a two-terminal electronic device, which is made by bringing p-type layer in close contact with n-type layer.

The electrode attached to the **p-side** of the junction is called the ***anode*** while the one attached to the **n-side** is called the ***cathode*** (Fig. 1a).

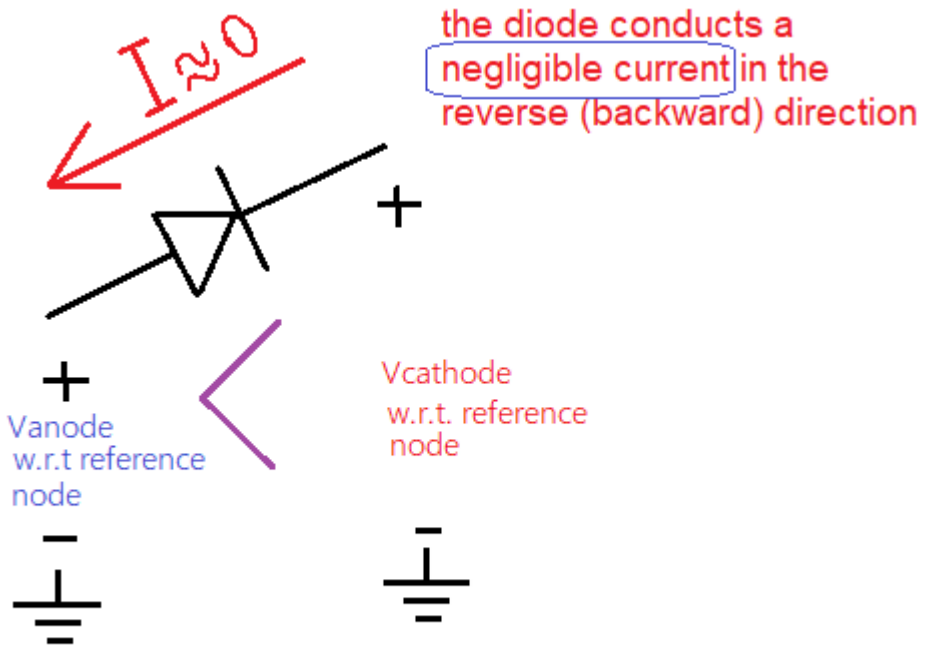
The **cathode** of a diode is usually *marked by a band* or a dot.



DIODE CONDITION TO CONDUCT CURRENT

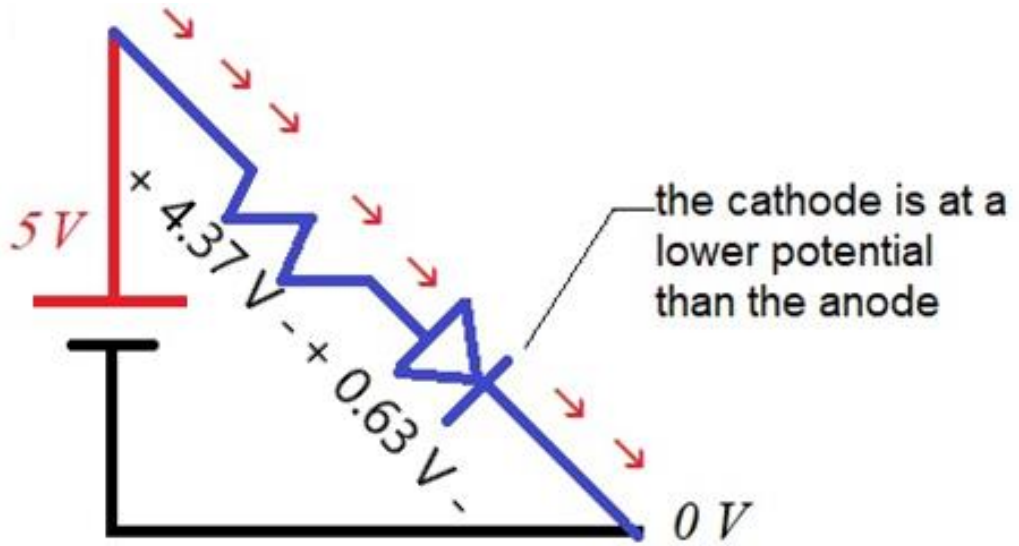


Forward-Biased Diode



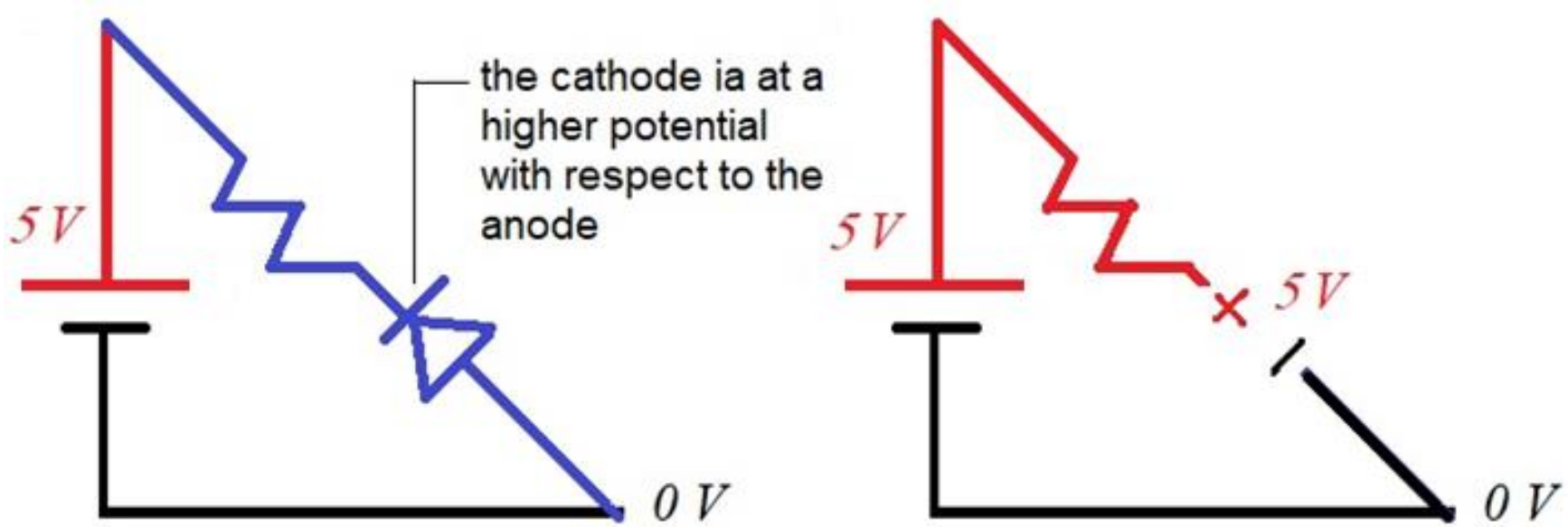
Reverse-Biased Diode

Forward Biasing a Diode



When a diode is put in a circuit so that it undergoes a voltage drop from anode to cathode, it becomes forward-biased and conducts forward current.

Reverse Biasing a Diode



When a diode is put in a circuit so that it undergoes a voltage drop from cathode to anode, it becomes reverse-biased and conducts no current;
it acts as an open circuit.

SUMMARY OF DIODE BEHAVIOUR

The diode conduction state mainly depends on the polarity of its [**anode**-to-**cathode**] terminal voltage.

To change its forward current 10 times, only about 0.1 V change in its terminal voltage is needed

A reverse-biased diode (*having the magnitude of its terminal voltage less than its breakdown voltage*) normally does not conduct $\Rightarrow$ can be modeled by an open circuit.

COMMON TYPES OF DIODES

High-power diodes are called *rectifier diodes*.

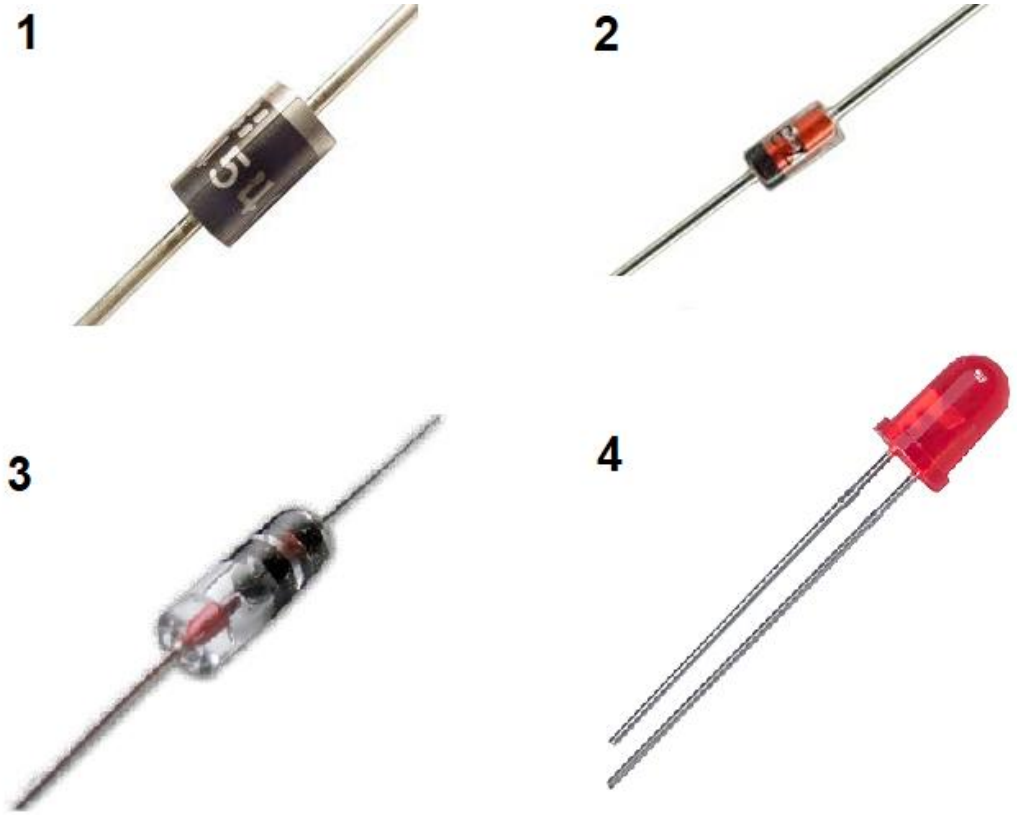
Diodes with high switching speeds are called *fast recovery* or *switching diodes*.

1- Rectifier diode.

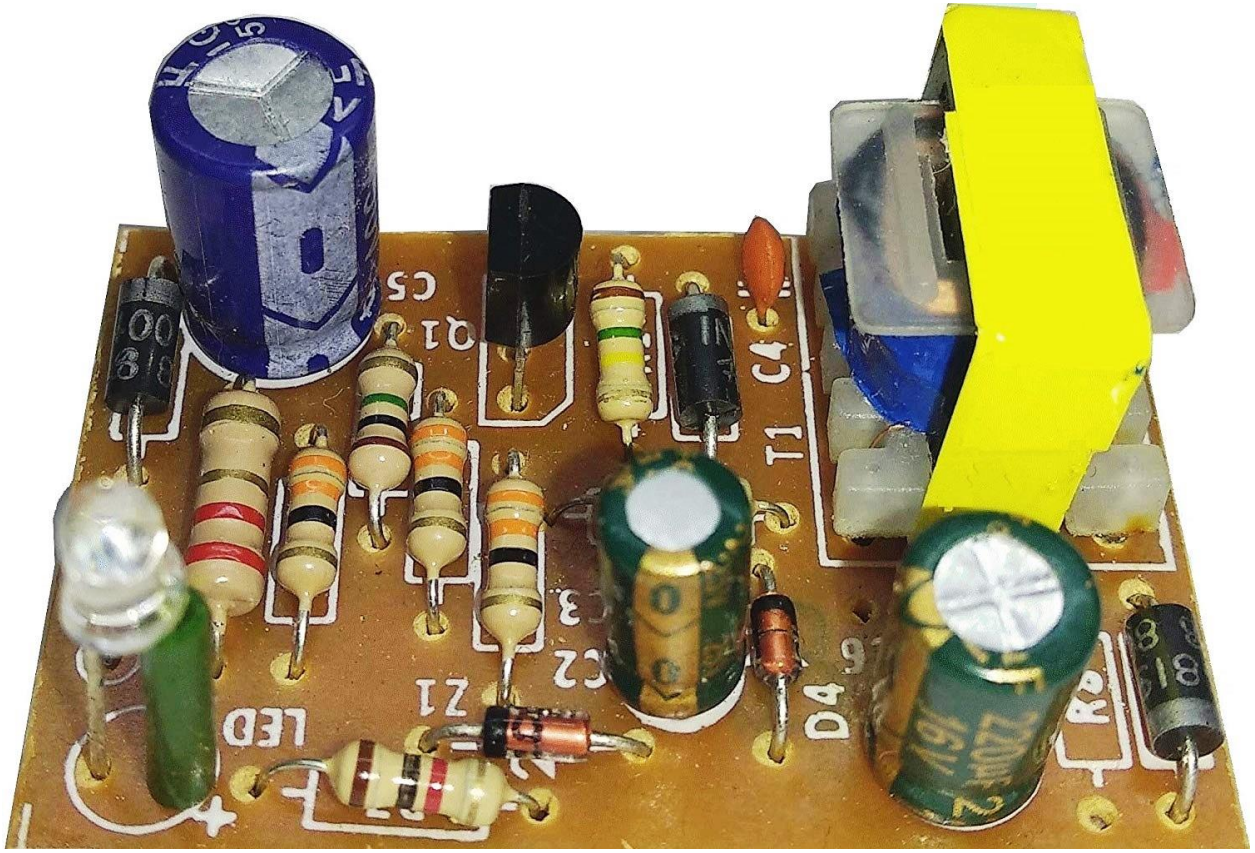
2- Zener diode which is made to handle reverse breakdown current safely.

3- Germanium diode.

4- An LED. It emits light while conducting a forward current.



Identify Three Types of Diodes in this Circuit Board



HOW TO TEST a DIODE USING a DMM

- Put your DMM on the diode-test mode.
- Record the open-circuit reading.
- Connect the diode to the DMM.
- If the DMM the reading does not change ⇒ the **cathode** should be the terminal connected to the **red jack (HI)** of the DMM.
- If the DMM displays a voltage drop, the **cathode** should be the terminal connected to the **black jack (COM or LO)** of the DMM.
- If the DMM displays similar readings for both directions, this implies that the diode is most probably defective.



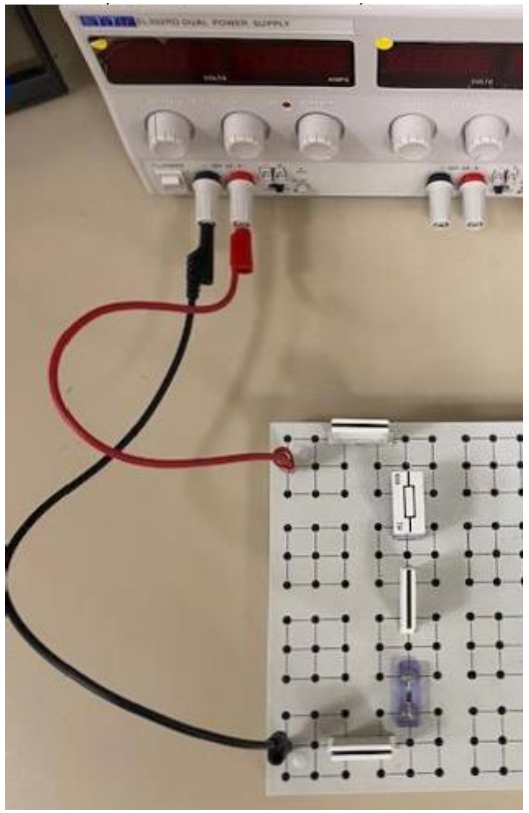
Procedure A Testing Common Types of Diodes

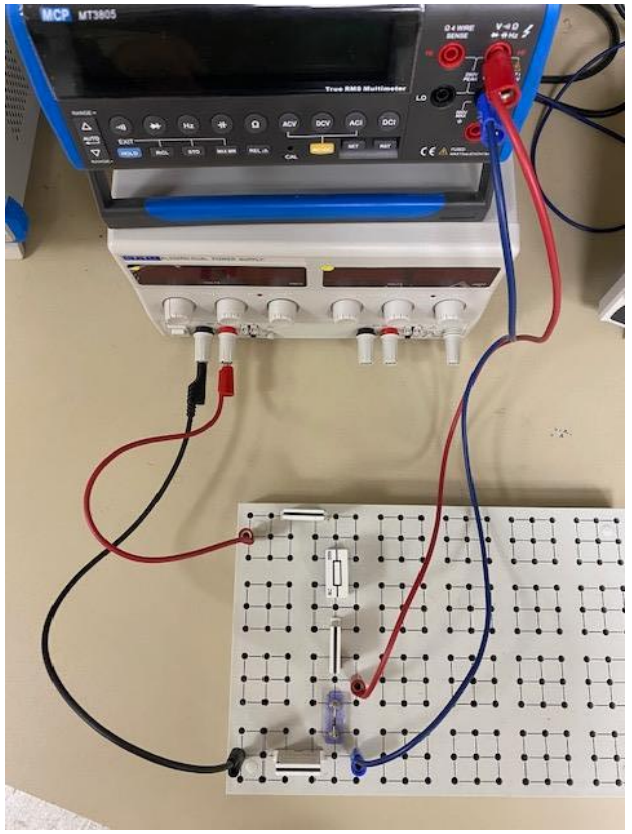


Table 1.1 Procedure A Testing Junction Diodes

DMM Test-Position Reading in Case of an Open Circuit:		
DMM Test-Position Reading in Case of a Reverse-Biased Diode:		
Type of diode	*Expected* Forward Voltage Drop	Measured *Forward Voltage* Drop *(Round off to 3 Decimal Places)*
Rectifier	~ 0.5 – 0.8 V (made of Si)	
Zener		
LED	~ 1.5 – 3.5 V	
Germanium	~ 0.2 – 0.4 V	

Procedure B Exploring the Behavior of Diode





Procedure B

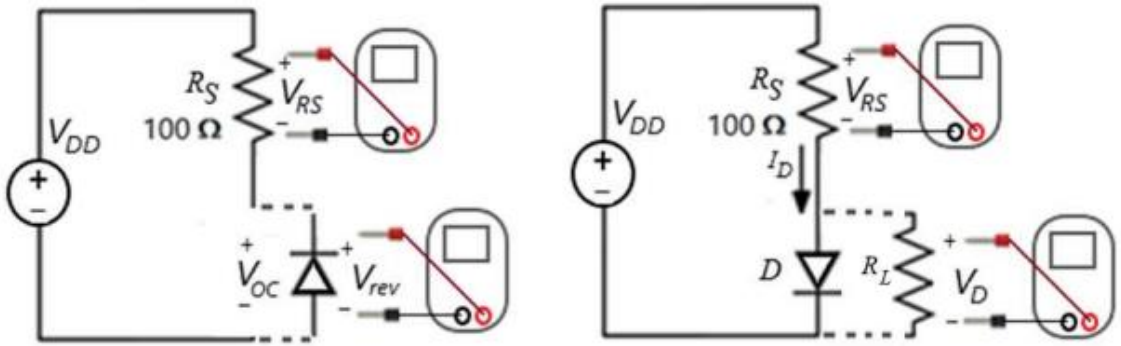
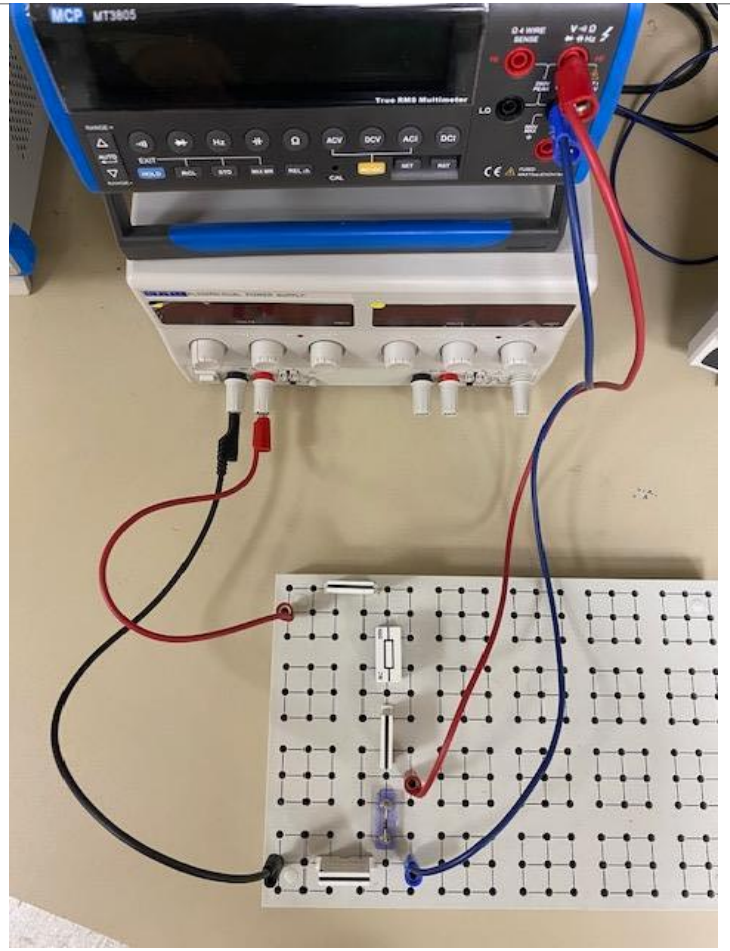


Table 1.2 Procedure B Measured Data *(Rounded off to 3 Decimal Places)*

Reverse Bias				
Step				
4, 5	Supply Voltage V_{DD} = 5 V			
	The corresponding open-circuit voltage V_{OC} =			
	The diode's reverse voltage drop V_{rev} =			
Forward Bias				
Step	Arrangement	I_D (mA)	V_D (V)	Change in V_D
7	Adjust V_{DD} so that V_{RS} = 0.1 V	1		
8	Adjust V_{DD} so that V_{RS} = 1 V	10		
9	Adjust V_{DD} so that V_{RS} = 10 V	100		



REFERENCES

T. A. Almadhi, EE 312 Lab Manual:

https://lms.ksu.edu.sa/bbcswebdav/users/talmadhi/EE312/EE312.pdf